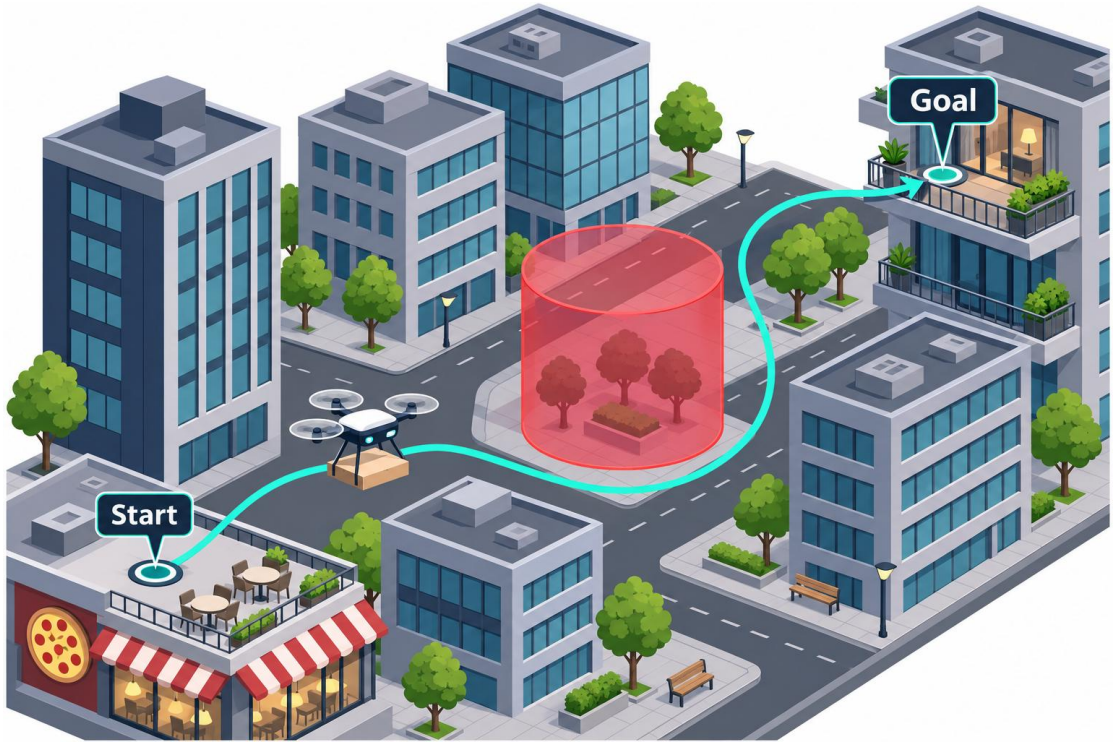
EEEE4138 Aerial Robotics
Topic 6

Trajectory Generation I

Route planning and smooth motion

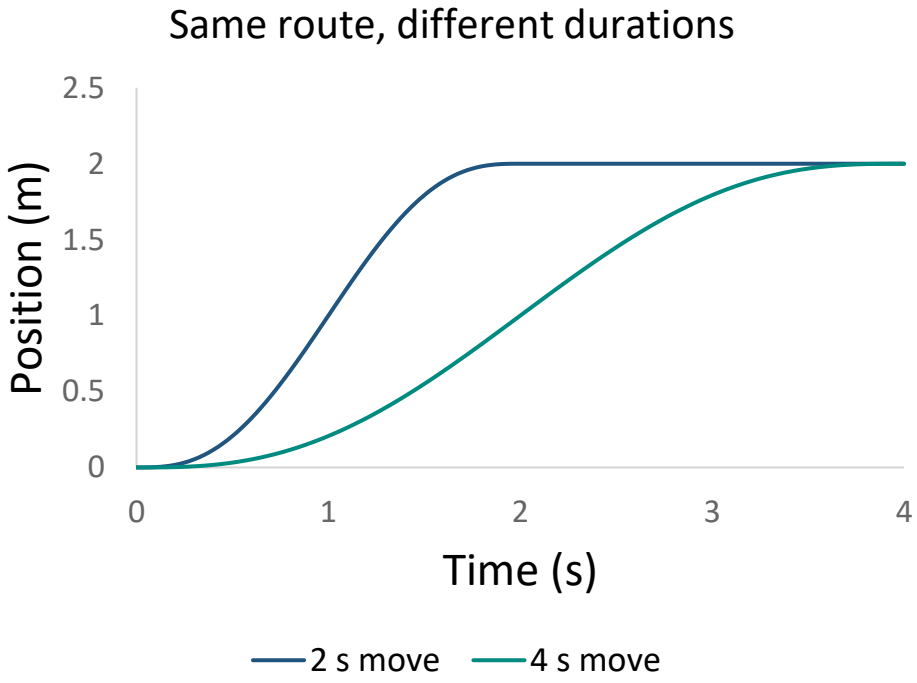


A delivery drone needs a safe route and a timetable.

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A path needs a timetable

A path specifies the geometric route.
A trajectory specifies position at each time.
The move can take 2 s or 4 s.



Both references travel along the same straight path from 0 to 2 m.

Planning algorithm families

Family	Main role	Examples
Search	Explore states to reach a goal.	DFS, BFS, greedy best-first, bidirectional search, A*
Path finding	Find a route using a stated cost.	Dijkstra, Bellman–Ford, Floyd–Warshall, bidirectional Dijkstra
Path generation	Construct routes or fit geometric curves.	RRT, PRM, cubic splines, Bézier curves
Trajectory generation	Specify motion over time with derivative conditions.	Polynomial, minimum-jerk/snap, timed Bézier and spline trajectories

The families overlap. RRT and PRM do not automatically return smooth trajectories.

Search methods: concept and use

Algorithm	How it selects work	Use or limitation
Depth-first search (DFS)	Follow one branch, then backtrack.	Reachability and exploration. No shortest-path guarantee.
Breadth-first search (BFS)	Expand one depth level at a time.	Fewest edges. Minimum-cost guarantee for equal edge costs.
Greedy best-first	Expand smallest heuristic h(n).	Goal-directed, but can miss a cheaper route.
Bidirectional search	Search from start and goal.	Can reduce exploration. Needs valid reverse moves and stopping.
A* (A-star)	Expand smallest f(n) = g(n) + h(n).	Optimal on a finite nonnegative graph with a consistent heuristic.

g(n): cost so far. h(n): estimated remaining cost. A* with h = 0 becomes Dijkstra.

Weighted path finding: concept and use

Algorithm	Core idea	Use or condition
Dijkstra	Expand the smallest accumulated cost.	Shortest paths with nonnegative edge costs.
Bellman–Ford	Repeatedly relax all edges.	Allows negative edges. Detects reachable negative cycles.
Floyd–Warshall	Try every vertex as an intermediate.	All-pairs shortest paths. Finite results require no negative cycles.
Bidirectional Dijkstra	Search by cost from both ends.	Single start–goal query. Needs a valid bound-based stopping rule.

A shortest route is optimal for its chosen cost and graph, not automatically for the aircraft.

Sampling planners: concept and use

Algorithm	Construction	Typical role
Rapidly exploring random tree (RRT)	Sample a point, extend a tree toward it, reject colliding extensions.	Explore free space for a start–goal query.
Probabilistic roadmap (PRM)	Sample free configurations and connect collision-free neighbours into a roadmap.	Reuse a roadmap for multiple route queries.

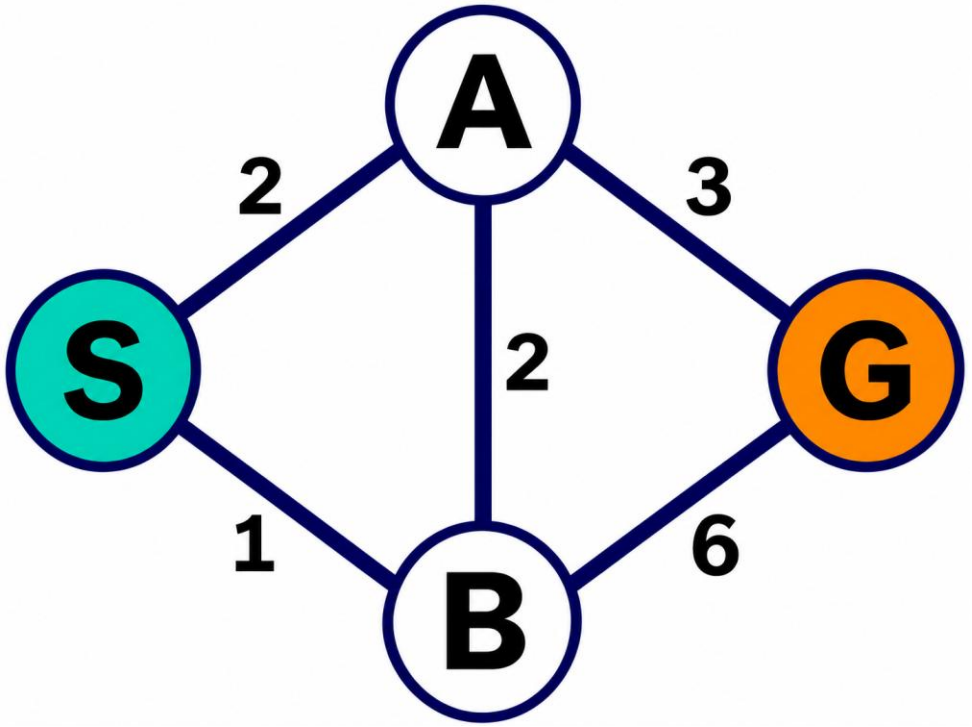
Their output is a route through connected configurations. Timing and smoothing may follow.

Worked route: cost matters

Undirected edges carry abstract costs.

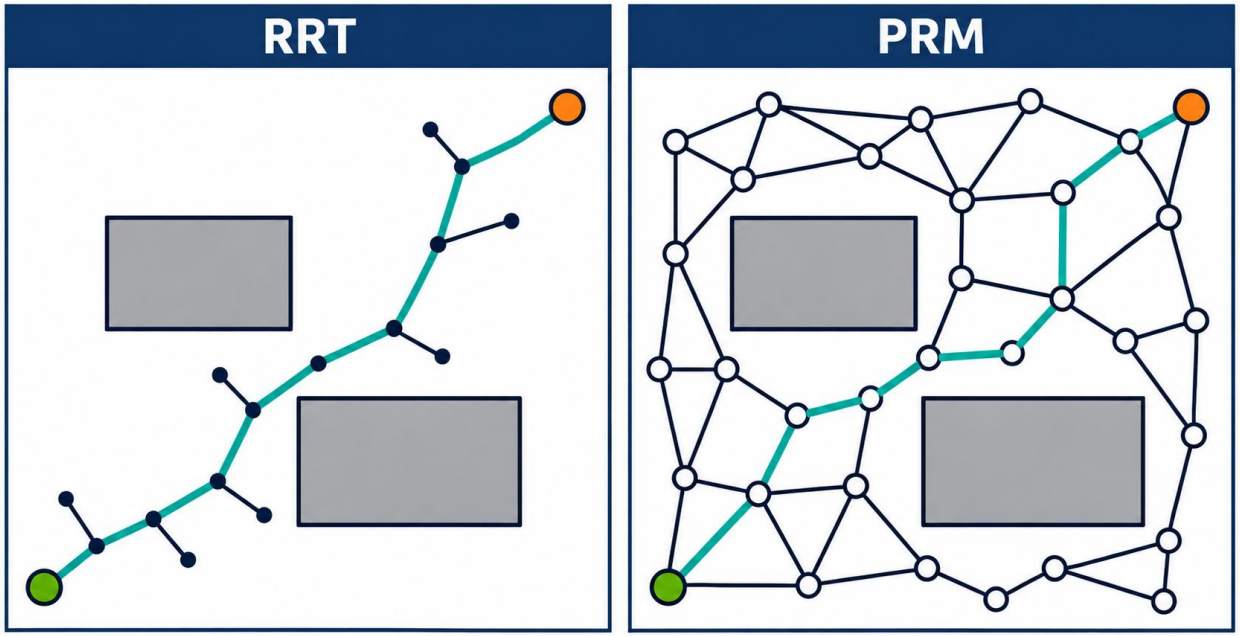
Find the minimum-cost route from S to G.

Compare the number of edges with total cost.



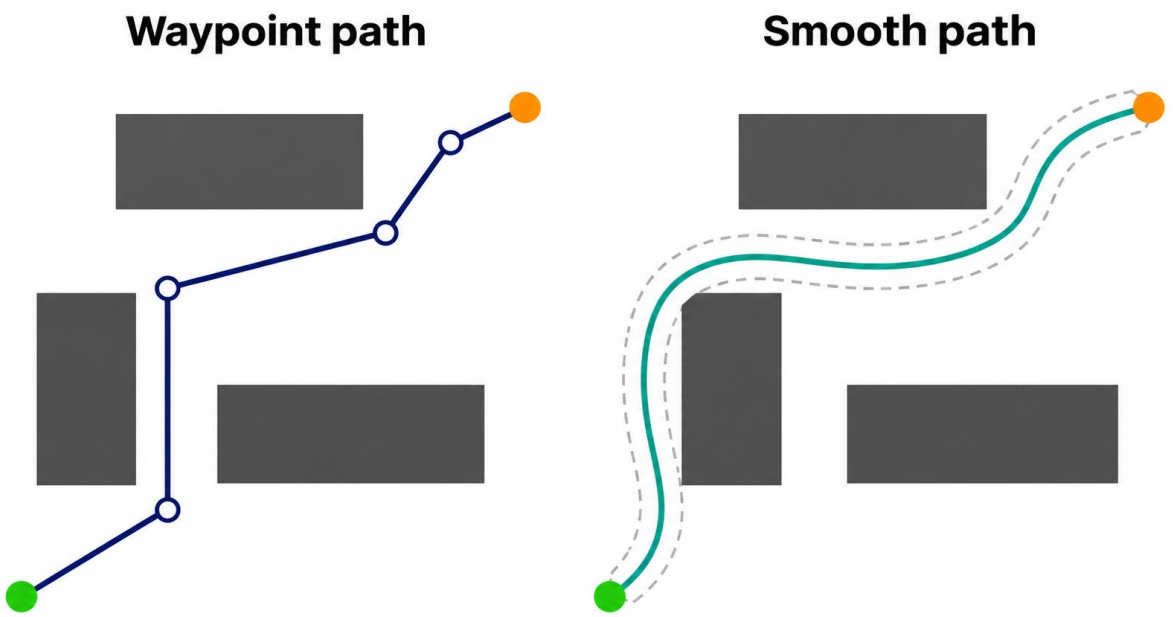
Drawing lengths are schematic. Use the edge labels as costs.

RRT and PRM: tree versus roadmap



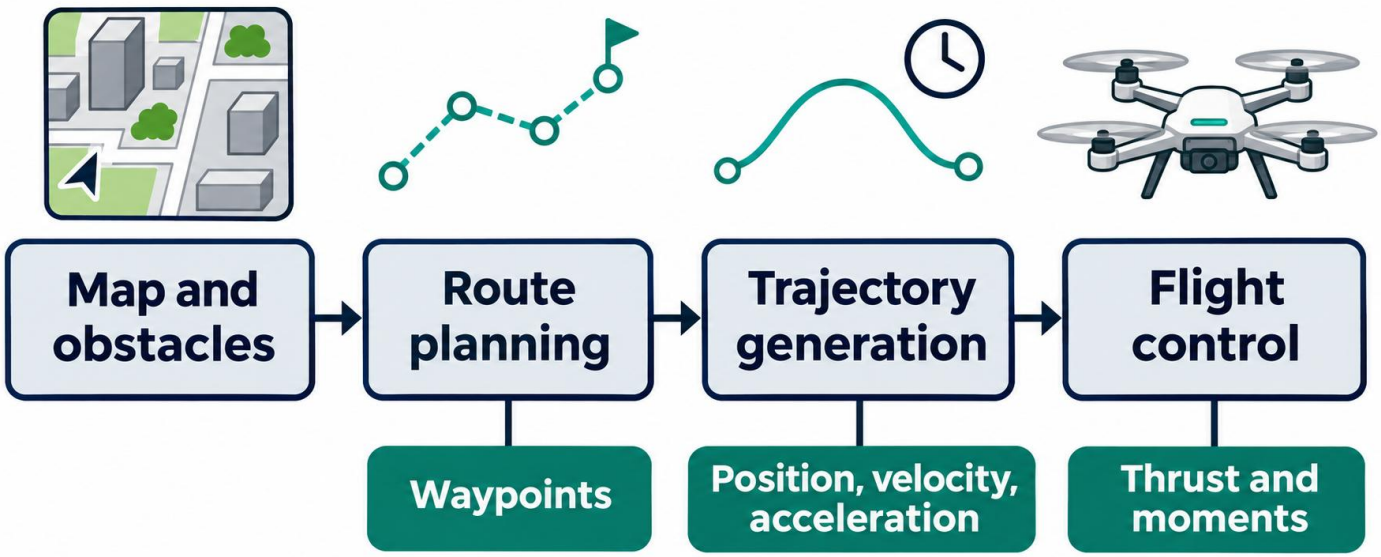
RRT grows a connected tree by checked extensions.
PRM builds a network of checked local connections.

A route can still contain sharp corners



Splines and Bézier curves can smooth route geometry.
A sharp corner at nonzero speed requires an abrupt velocity chang

Planning, trajectory generation and control



This is one possible workflow, not a compulsory sequence.
The planner supplies a route or waypoint set.

What information defines the problem?

Initial and final position

Required endpoint velocity and acceleration

Duration and any intermediate waypoints

A smoothness objective and physical limits

Position and its derivatives

Position x [m] and velocity $v = dx/dt$ [m/s]

Acceleration $a = d^2x/dt^2$ [m/s²]

Jerk $j = d^3x/dt^3$ [m/s³]

Jerk describes how quickly acceleration changes.

Boundary conditions describe the endpoints

Endpoint quantity	Rest-to-rest cubic	Quintic joined to holds
Position	$x(0) = x_o$, $x(D) = x_f$	$x(0) = x_o$, $x(D) = x_f$
Velocity	$v(0) = v(D) = 0$	$v(0) = v(D) = 0$
Acceleration	Not prescribed	$a(0) = a(D) = 0$

Zero endpoint velocity does not imply zero endpoint acceleration.

Polynomial degree and available coefficients

Degree n has n + 1 coefficients.

Two position conditions can determine a linear polynomial.

Four position/velocity conditions can determine a cubic.

Six position/velocity/acceleration conditions can determine a quintic.

Local time: reset the stopwatch

Start at $t_0 = 5$ s and finish at $t_f = 8$ s

These are readings on the global clock.

$D = t_f - t_0 = 8 - 5 = 3$ s

D is the duration of this movement.

At $t = 6$ s: $\tau = t - t_0 = 6 - 5 = 1$ s

τ is the time elapsed since this movement started.

The movement starts at $\tau = 0$ and finishes at $\tau = D$.

Normalised time: the fraction of duration elapsed

$s = \tau/D$	Divide elapsed time by the movement duration.
For $D = 3$ s and $\tau = 1$ s: $s = 1/3$	One third of the allotted time has elapsed.
Start: $s = 0$. Finish: $s = 1$.	The same interval 0 to 1 works for any duration.

Here s is dimensionless. It is a time fraction, not a distance fraction.

Linear position: a constant-speed example

$x_0 = 0$ m, $x_f = 2$ m, $D = 2$ s

Travel 2 metres in 2 seconds.

$x(\tau) = x_0 + [(x_f - x_0)/D]\tau$
$= (1 \text{ m/s})\tau$

Position increases by 1 metre each second.

$v = dx/d\tau = 1$ m/s
$a = dv/d\tau = 0$ m/s²

These derivatives apply inside the movement.

A straight line on a position–time plot means constant velocity.

The problem at the start and finish

Before the move: v = 0

A stationary hold has constant position.

During the move: v = 1 m/s

The linear formula requests an immediate speed change.

After the move: v = 0

Stopping also requires an immediate speed change.

We need a position formula whose velocity can start and finish at zero.

Cubic position: differentiate term by term

$x = c_0 + c_1 T + c_2 T^2 + c_3 T^3$

Four coefficients are available to choose.

$v = c_1 + 2c_2 T + 3c_3 T^2$

Use $d(T^n)/dT = nT^{n-1}$. The constant c_0 disappears.

$a = 2c_2 + 6c_3 T$

Differentiate the velocity once more.

We choose c_0 to c_3 so that the endpoint requirements are satisfied.

Cubic: the start gives two coefficients

$x(0) = c_0 = x_0$

At $\tau = 0$, every term containing τ disappears.

$v(0) = c_1 = v_0$

The starting velocity fixes c_1.

$x = x_0 + v_0\tau + c_2\tau^2 + c_3\tau^3$

Only c_2 and c_3 remain unknown.

For a start from rest, $v_0 = 0$, so $c_1 = 0$.

Cubic: the finish gives two equations

$x_f = x_0 + v_0 D + c_2 D^2 + c_3 D^3$

Evaluate the position at the finishing time $\tau = D$.

$v_f = v_0 + 2c_2 D + 3c_3 D^2$

Evaluate velocity at the same finishing time.

For $v_0 = v_f = 0$:
$c_2 D^2 + c_3 D^3 = \Delta x$
$2c_2 D + 3c_3 D^2 = 0$

Let $\Delta x = x_f - x_0$. These two equations determine c_2 and c_3.

Two unknown coefficients remain, and we have two independent equations.

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Cubic: eliminate one unknown

$2c_2D + 3c_3D^2 = 0$
$c_2 = -(3/2)c_3D$

Rearrange the final-velocity equation.

$-(3/2)c_3D^3 + c_3D^3 = \Delta x$
$-(1/2)c_3D^3 = \Delta x$

Substitute c_2 into the final-position equation.

$c_3 = -2\Delta x/D^3$
$c_2 = 3\Delta x/D^2$

Solve for c_3, then substitute back for c_2.

The coefficient formulas come from the endpoint conditions.

Worked cubic: calculate the coefficients

$x_0 = 0$ m, $x_f = 2$ m, $D = 2$ s
$\Delta x = 2 - 0 = 2$ m

Require zero starting and finishing velocity.

$c_0 = 0$; $c_1 = 0$
$c_2 = 3(2)/2^2 = 1.50$ m/s²

The starting conditions and displacement fix these values.

$c_3 = -2(2)/2^3 = -0.500$ m/s³
$x(\tau) = 1.50\tau^2 - 0.500\tau^3$

Insert the coefficients into the position polynomial.

Use seconds for τ. Each term in x has units of metres.

Worked cubic: check the requested finish

$v = 3\tau - 1.50\tau^2$
$a = 3 - 3\tau$

Differentiate the numerical position formula.

At $\tau = 2$ s:
$x = 1.50(4) - 0.500(8) = 2$ m
$v = 3(2) - 1.50(4) = 0$

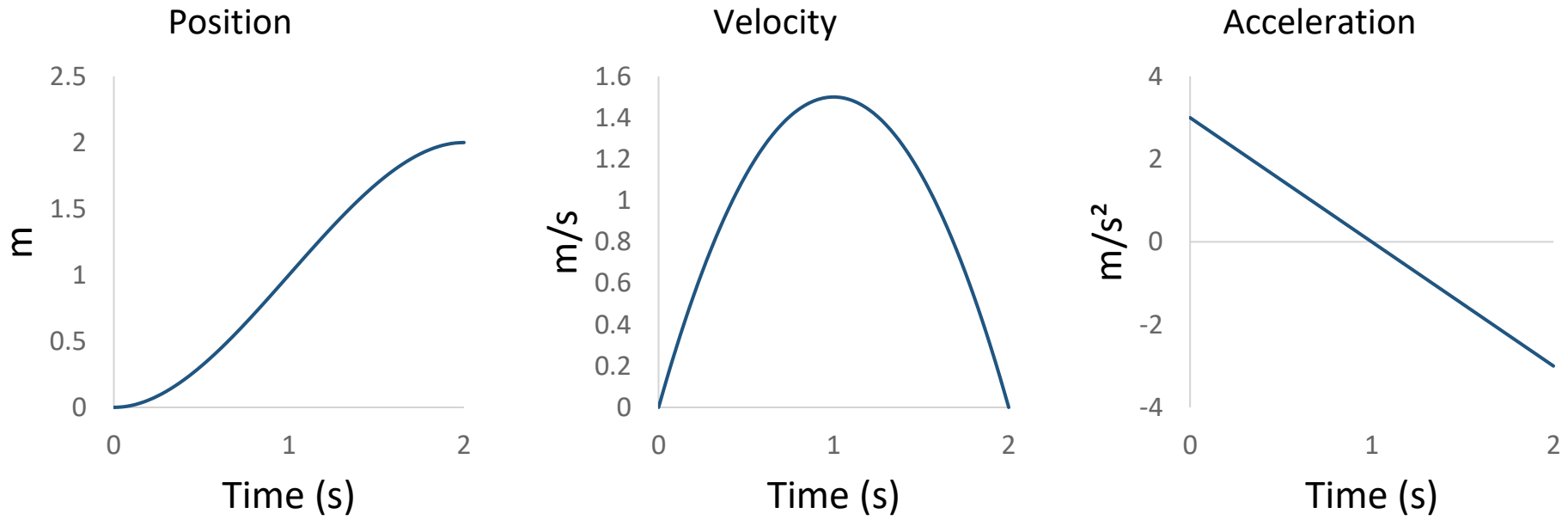
The finishing position and velocity are correct.

$a(0) = +3$ m/s²
$a(2) = -3$ m/s²

Endpoint acceleration was not constrained to zero.

The cubic meets four endpoint conditions, but it does not match hold acceleration.

Cubic motion: what the graphs show



Example: a 2 m move in 2 s, with zero endpoint velocity.

Position and velocity match the holds. Acceleration jumps at the joins.

Try the same method

Move from 1 m to 4 m in 3 s

Require zero velocity at the start and finish.

Find c_0, c_1, c_2 and c_3

Use $\Delta x = x_f - x_0$ and the cubic formulas.

Find velocity at $\tau = 1.50$ s

Also check whether acceleration joins a stationary hold.

Use local time from zero, and show the substitutions.

Cubic practice: the worked answer

$\Delta x = 4 - 1 = 3$ m
$c_0 = 1$; $c_1 = 0$

The starting position is 1 m, not zero.

$c_2 = 3(3)/3^2 = 1$
$c_3 = -2(3)/3^3 = -2/9$

Insert displacement and duration into the formulas.

$x = 1 + \tau^2 - (2/9)\tau^3$
$v = 2\tau - (2/3)\tau^2$

Differentiate position to obtain velocity.

The coefficient units are m, m/s, m/s² and m/s³, respectively.

Cubic practice: midpoint and acceleration

$v(1.50) = 2(1.50) - (2/3)(1.50)^2$
$= 3 - 1.50 = 1.50$ m/s

Substitute the midpoint into the velocity formula.

$a = dv/d\tau = 2 - (4/3)\tau$

Differentiate velocity once more.

$a(0) = +2$ m/s²
$a(3) = 2 - 4 = -2$ m/s²

Neither endpoint matches a stationary hold's zero acceleration.

Zero endpoint velocity does not imply zero endpoint acceleration.

Quintic motion: six endpoint conditions

Start: $x(0) = x_0$, $v(0) = 0$, $a(0) = 0$

Specify position, velocity and acceleration at the start.

Finish: $x(D) = x_f$, $v(D) = 0$, $a(D) = 0$

Specify the same three quantities at the finish.

$$x = c_0 + c_1 T + c_2 T^2 + c_3 T^3 + c_4 T^4 + c_5 T^5$$

A quintic has six coefficients for these six conditions.

The extra conditions allow position, velocity and acceleration to match holds.

Quintic: apply the starting conditions

$x(0) = c_0 = x_0$

Starting position determines c_0.

$v(0) = c_1 = 0$
$a(0) = 2c_2 = 0$, so $c_2 = 0$

Starting at rest with zero acceleration fixes c_1 and c_2.

$x = x_0 + c_3T^3 + c_4T^4 + c_5T^5$

Only c_3, c_4 and c_5 remain to be found.

Three starting conditions have already determined three coefficients.

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Quintic: apply the finishing conditions

$c_3D^3 + c_4D^4 + c_5D^5 = \Delta x$

Final position gives this equation, with $\Delta x = x_f - x_0$.

$3c_3D^2 + 4c_4D^3 + 5c_5D^4 = 0$

Differentiate once and set v(D) = 0.

$6c_3D + 12c_4D^2 + 20c_5D^3 = 0$

Differentiate twice and set a(D) = 0.

We now have three equations for three unknown coefficients.

Scaling removes the powers of duration

$b_3 = c_3D^3$, $b_4 = c_4D^4$, $b_5 = c_5D^5$

These are temporary algebra variables.

$b_3 + b_4 + b_5 = \Delta x$
$3b_3 + 4b_4 + 5b_5 = 0$

Multiply v(D)=0 by D. For example: $3c_3D^2 \times D = 3b_3$.

$6b_3 + 12b_4 + 20b_5 = 0$

Multiply a(D)=0 by D^2. For example: $6c_3D \times D^2 = 6b_3$.

All b values have units of displacement. The equations are easier to solve.

Quintic: eliminate b_3

Equation 2 − 3 × equation 1:
$b_4 + 2b_5 = -3\Delta x$

The b_3 terms cancel because $3 - 3 = 0$.

Equation 3 − 2 × equation 2:
$4b_4 + 10b_5 = 0$

Again, b_3 cancels because $6 - 6 = 0$.

$b_4 = -(10/4)b_5 = -2.5b_5$

Use the second reduced equation to express b_4 in terms of b_5.

Only two unknowns remain in the reduced equations.

Quintic: recover the coefficients

$-2.5b_5 + 2b_5 = -3\Delta x$
$b_5 = 6\Delta x$, $b_4 = -15\Delta x$

Substitute into $b_4+2b_5=-3\Delta x$.

$b_3 = \Delta x - b_4 - b_5 = 10\Delta x$

Use the original position equation.

$c_3 = 10\Delta x/D^3$
$c_4 = -15\Delta x/D^4$, $c_5 = 6\Delta x/D^5$

Divide by the duration powers used in the b definitions.

Together with $c_0=x_0$ and $c_1=c_2=0$, the quintic is fully determined.

One reusable quintic timing function

$$x = x_0 + \Delta x[10(\tau/D)^3 - 15(\tau/D)^4 + 6(\tau/D)^5]$$

Substitute the coefficients into the polynomial.

$$s = \tau/D;\ S(s) = 10s^3 - 15s^4 + 6s^5$$

Collect the time dependence in one function.

$$x = x_0 + \Delta x\ S(s)$$

S(s) gives the fraction of displacement completed.

s is the fraction of time elapsed. S(s) is the fraction of displacement completed.

Velocity: use the chain rule

$s = (t - t_0)/D$, so $ds/dt = 1/D$

D is constant for this segment.

$dx/dt = \Delta x\ [dS/ds]\ [ds/dt]$

S depends on time through s. Differentiate both links.

$v = (\Delta x/D)\ S'(s)$

The prime means differentiation with respect to s.

Do not confuse S′(s) with physical velocity. Multiply it by Δx/D.

Acceleration and jerk: another factor of 1/D

$a = d[(\Delta x/D)S']/dt$
$= (\Delta x/D^2)S''(s)$

Differentiate again. The chain rule adds another 1/D.

$j = da/dt = (\Delta x/D^3)S'''(s)$

A third derivative adds a third factor of 1/D.

Double D for the same path and shape:
$v \div 2$, $a \div 4$, $j \div 8$

Compare corresponding normalised times s.

A longer duration reduces the derivative magnitudes for this same motion shape.

Differentiate S using the power rule

$S = 10s^3 - 15s^4 + 6s^5$
$S' = 30s^2 - 60s^3 + 30s^4$

Multiply each coefficient by its power, then reduce the power by one.

$S'' = 60s - 180s^2 + 120s^3$

Differentiate S' to obtain the normalised acceleration shape.

$S''' = 60 - 360s + 360s^2$

Differentiate S'' to obtain the normalised jerk shape.

These derivatives are with respect to s. Duration scaling comes afterwards.

Halfway through the quintic

At s = 0.5:

$S = 10(0.5)^3 - 15(0.5)^4 + 6(0.5)^5$

$= 1.25 - 0.9375 + 0.1875 = 0.5$

Halfway in time is also halfway in displacement for this symmetric move.

$S'(0.5) = 7.5 - 7.5 + 1.875$

$= 1.875$

This is a dimensionless derivative value.

$S''(0.5) = 30 - 45 + 15 = 0$

Physical acceleration is also zero at the midpoint.

To obtain physical velocity, multiply 1.875 by Δx/D.

Worked quintic: calculate the coefficients

$x_0 = 0$ m, $x_f = 2$ m, $D = 2$ s
$\Delta x = 2$ m

Require zero endpoint velocity and acceleration.

$c_3 = 10(2)/2^3 = 2.50$
$c_4 = -15(2)/2^4 = -1.875$

The denominator powers are 8 and 16.

$c_5 = 6(2)/2^5 = 0.375$
$x = 2.50\tau^3 - 1.875\tau^4 + 0.375\tau^5$

The final denominator is 32.

Coefficient units are m/s^3, m/s^4 and m/s^5 for c_3, c_4 and c_5.

Worked quintic: evaluate the midpoint

At $\tau = 1$ s: $s = 1/2 = 0.5$
$x = x_0 + \Delta x S = 0 + 2(0.5) = 1$ m

Use the normalised values from the previous calculation.

$v = (\Delta x / D) S'$
$= (2/2)(1.875) = 1.875$ m/s

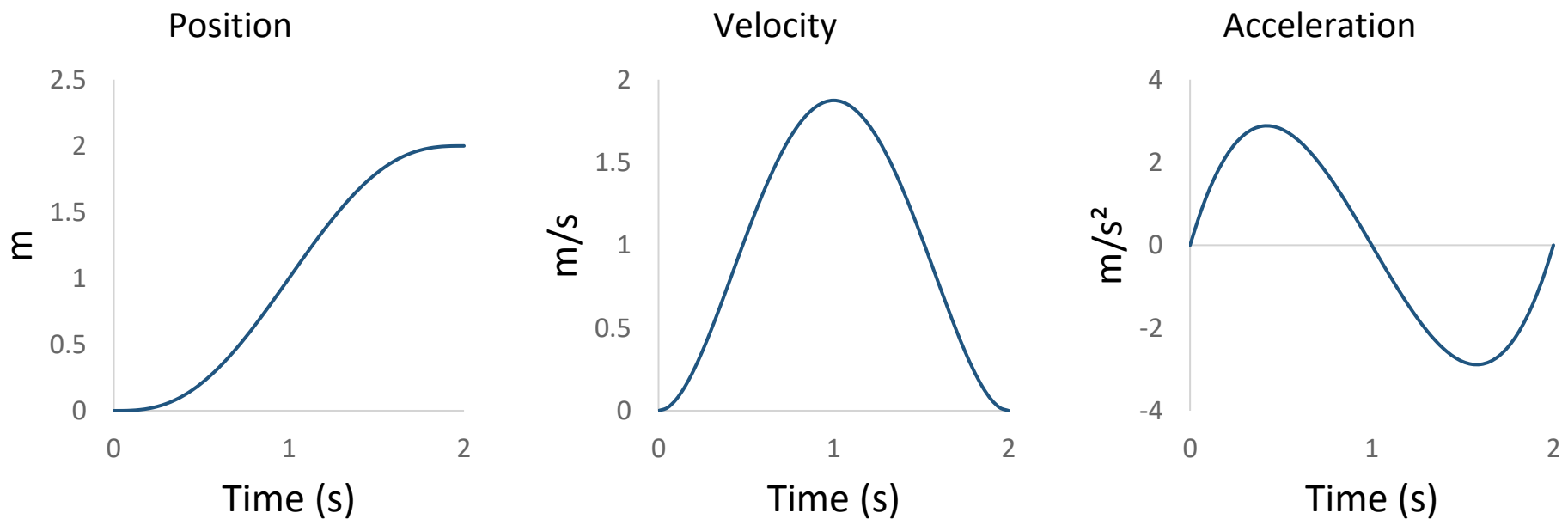
Displacement divided by duration converts S' to velocity.

$a = (\Delta x / D^2) S''$
$= (2/4)(0) = 0$ m/s²

Zero acceleration at this instant does not mean zero velocity.

The aircraft reference is moving fastest at the midpoint of this example.

Quintic motion: smooth joins to holds



Example: a 2 m move in 2 s, with zero endpoint velocity.

Position, velocity and acceleration match the holds. Jerk need not match.

Minimum jerk: what is being minimised?

$j(t) = da/dt$

Jerk measures how rapidly acceleration changes.

$J = \int_0^D [j(t)]^2 \, dt$

Square jerk, then accumulate it over the movement.

Fix D and endpoint x, v and a

Compare candidate motions with the same timing and endpoint requirements.

For this unconstrained problem, the minimum-jerk solution is a quintic.

The derivative cost determines the polynomial

Minimise $\int v^2 dt$ Fix endpoint positions	The unconstrained solution is linear.
Minimise $\int a^2 dt$ Fix endpoint position and velocity	The unconstrained solution is cubic.
Minimise $\int j^2 dt$ Fix endpoint position, velocity and acceleration	The unconstrained solution is quintic.

These results use fixed duration and the stated endpoint conditions.

A vertical climb: keep the sign convention

$z_0 = -1$ m, $z_f = -4$ m, $D = 3$ s
$\Delta z = -4 - (-1) = -3$ m

Downwards is positive, so climbing makes z more negative.

$s = t/3$; $z_d = -1 - 3S(s)$

This example starts at global time $t_0=0$.

$\dot{z}_d = (-3/3)S' = -S'$
$\ddot{z}_d = (-3/3^2)S'' = -S''/3$

Apply the displacement and duration factors to the derivatives.

Negative vertical velocity means upward motion in this convention.

Vertical climb: check the midpoint and finish

At $t = 1.50$ s: $s = 0.5$
$z_d = -1 - 3(0.5) = -2.50$ m

The reference is halfway between the endpoint positions.

$\dot{z}_d = -1.875$ m/s; $\ddot{z}_d = 0$

It is moving upward with zero acceleration at this instant.

At $t = 3$ s: $S = 1$, $S' = S'' = 0$
$z_d = -4$ m, $\dot{z}_d = 0$, $\ddot{z}_d = 0$

The reference finishes at rest and matches a stationary hold.

Check position and derivatives separately. Each has its own physical meaning.

Before connecting the reference to the controller

Substitute $\tau = 0$ and $\tau = D$ into x, v and a	Confirm the endpoint values analytically.
Use $s = (t - t_0)/D$ and the correct 1/D factors	Check units and duration scaling in every derivative.
Before the move: $x = x_0$ After the move: $x = x_f$	For these stationary holds, set velocity and acceleration to zero.

Use the polynomial only during its designed interval. Plot references before tracking.

Recap: the reasoning behind the equations

Four endpoint x/v values give four equations	A cubic supplies four coefficients.
Six endpoint x/v/a values give six equations	A quintic supplies six coefficients and can match hold acceleration.
Every derivative through s adds a factor 1/D	The chain rule converts the normalised shape into physical motion.